

Callable statement FunctionsWhat

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What is CallableStatement?

- Interface in java.sql package
- Extends PreparedStatement
- Executes stored procedures using SQL escape syntax:
 - "{call procedure_name(?,?,?...?)}"
 - "{?=call function_name(?,?,...)}" (for functions with return value)

Important function of CallableStatement

1. Registering OUT parameters

- a. `registerOutParameter(int parameterIndex, int sqlType);`
Defines the type of data expected from the stored procedure.

2. Setting IN parameters

- Inherited from PreparedStatement:
- 1) `setInt(int index, int value);`
 - 2) `setString(int index, String value);`
 - 3) `setDate(int index, Date value);`

3. Retrieving OUT parameters

- After execution use getXXX() methods:
- 1) `getInt(int index)`
 - 2) `getString(int index)`
 - 3) `getDate(int index)`

4. Executing Stored procedures

- i. `execute()` -> Executes the stored procedure.
- ii. `executeUpdate()` -> For procedures that modify data.
- iii. `executeQuery()` -> For procedures returning a result set.

"{?=call function_name(?,?,...)}" (for functions with return value). This is slightly different from calling stored procedures because function return of value directly.

To call the function in JDBC, you can use the escape syntax:

```
{?=call function_name(?,?,...)}
```

?= -> placeholder for the return value

Function_name -> the name of the database function

(?,?,...) -> placeholders for input parameters

```
create or replace NONEDITIONABLE FUNCTION GETAVG  
(
```

```

    E_ID1 IN VARCHAR2
, E_ID2 IN VARCHAR2
) RETURN NUMBER AS
sal1 NUMBER;
sal2 NUMBER;
BEGIN
    select basic_salary INTO sal1 FROM employee where EMP_ID = e_id1;
    select basic_salary INTO sal2 FROM employee where EMP_ID = e_id2;
    return (sal1 + sal2)/2;
END;
/

```

```

package MyJDBCOne;
import java.sql.CallableStatement;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Types;
public class StoredProcedureFunctionDemo1 {
    public static void main(String[] args) {
        String dbUrl = "jdbc:oracle:thin:@//localhost:1521/FREE";
        String username = "sys as sysdba";
        String password = "Calcutta1983";
        try {
            Connection con = DriverManager.getConnection(dbUrl, username,
password);
            CallableStatement cst = con.prepareCall("{?= call getAvg(?, ?)}");
            cst.setString(2, "E001");
            cst.setString(3, "E002");
            cst.registerOutParameter(1, Types.FLOAT);
            cst.execute();
            System.out.println("Average salary: " + cst.getFloat(1));
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}

```

Output:

Average salary: 58900.0

Functions are exactly same as procedures except that functions has a return statement directly. Procedure can also return values indirectly in the form of OUT parameters. Usually we can. Use procedure to define business logic and we can use functions to perform some calculations like getAverage(), getMax() etc.